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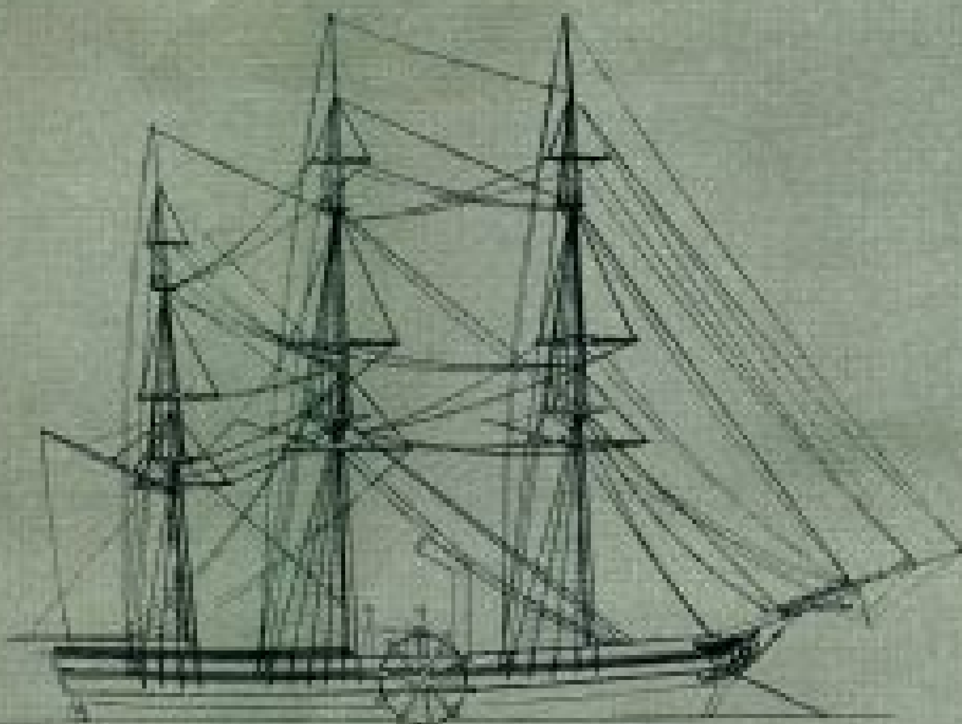

Paper 21, pages 61–80, from

CONTRIBUTIONS FROM THE MUSEUM OF HISTORY AND TECHNOLOGY

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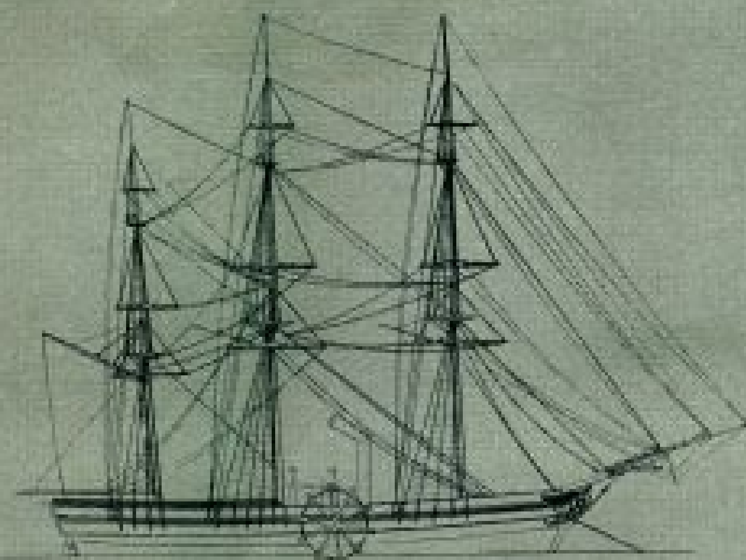

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Contributions from The Museum of History and Technology:

Paper 21

The Pioneer Steamship Savannah:

A Study for a Scale Model

Howard I. Chapelle

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The Pioneer Steamship SAVANNAH: *A Study for a Scale Model*

The original plans of the pioneer transatlantic steamer Savannah *no longer exist, and many popular representations of the famous vessel have been based on a 70-year-old model in the United States National Museum. This model, however, differs in several important respects from contemporary illustrations.*

To correct these apparent inaccuracies in a new, authentic model, a reconstruction of the original plans was undertaken, using as sources the ship's logbook and customhouse description, a French report on American steam vessels published in 1823, and Russian newspaper accounts contemporary with the Savannah's *visit to St. Petersburg on her historic voyage of 1819. The development of this research and the resulting information in terms of her measurements and general description are related here.*

THE AUTHOR: *Howard I. Chapelle is curator of transportation in the United States National Museum, Smithsonian Institution.*

The United States National Museum has in its watercraft collection a rigged scale model purported to be of the pioneer transatlantic steamer *Savannah*. For many years this model was generally accepted as being a reasonably accurate representation and was the basis for countless illustrations. Curiously enough, the model (USNM 160364) does not agree with the published catalog description[1] as to the side paddle wheels. Neither does it agree with the material in the Marestier report,[2] which is accepted as the only source for a contemporary picture of the *Savannah*.

The recent naming of an atomic-powered ship in honor of the famous steamer greatly increased popular interest in the pioneer ship and its supposed model. Consequently, the National Museum undertook the research necessary to correct or replace the existing model. This research has been carried out by the staff of the Museum's transportation division with the aid of Frank O. Braynard of the American Merchant Marine Institute, Eugene S. Ferguson, curator of mechanical and civil engineering at the Museum, and others.

The *Savannah* crossed from Savannah, Georgia, to Liverpool, England, in the period May 22 to June 20, 1819; and proceeded to the Baltic, where she entered at St. Petersburg (now Leningrad), Stockholm, and a few other ports. On her return she reached Savannah on November 30, and on December 3 she sailed for Washington, D.C., arriving on December 16. Her original logbook now on exhibition in the Museum,[3] covers the period between March 28, 1819, when she first left New York for Savannah, to December 1819 when she was at Washington.

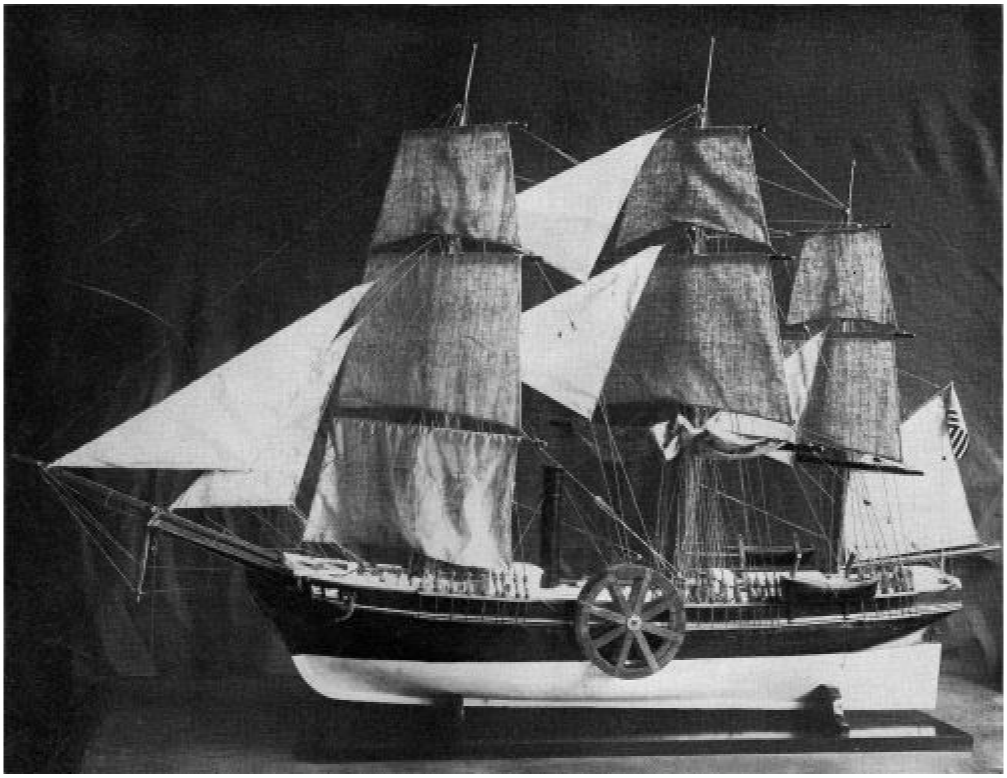

Figure 1.—Old model of the *Savannah*, built under the supervision of Captain Collins. This model has been removed from exhibition in the United States National Museum because of inaccuracies. (*USNM* 160364; *Smithsonian photo* 14355.)

The old model (fig. 1) was built about 1890–1892 by Lawrence Jenson, a master shipwright and model builder of Gloucester and Rockport, Massachusetts, under the supervision of Capt. Joseph Collins of the U.S. Fish Commission. Notes in the records of the Museum's transportation division show that the research for this model was done by Captain Collins through use of an unidentified lithograph, printed after the transatlantic voyage, and what then could be learned about American sailing ships contemporary with the *Savannah*. In these notes the complaint is made that no contemporary representation of the steamship had then been found.

The old, inaccurate model, built to the scale of one-half inch to the foot, represents an auxiliary, side-wheel, ship-rigged steamer. The model scale measurements are about 120 feet in over-all length, 29 feet in beam, and 13 feet 6 inches depth in hold. The tonnage is stated on the exhibit card to have been about 350 tons, old measurement. The model has crude wooden side paddles of the radial type, a tall straight smokestack between fore and main masts, a small deckhouse forward of the stack, a raised quarter-deck, and a round stern.

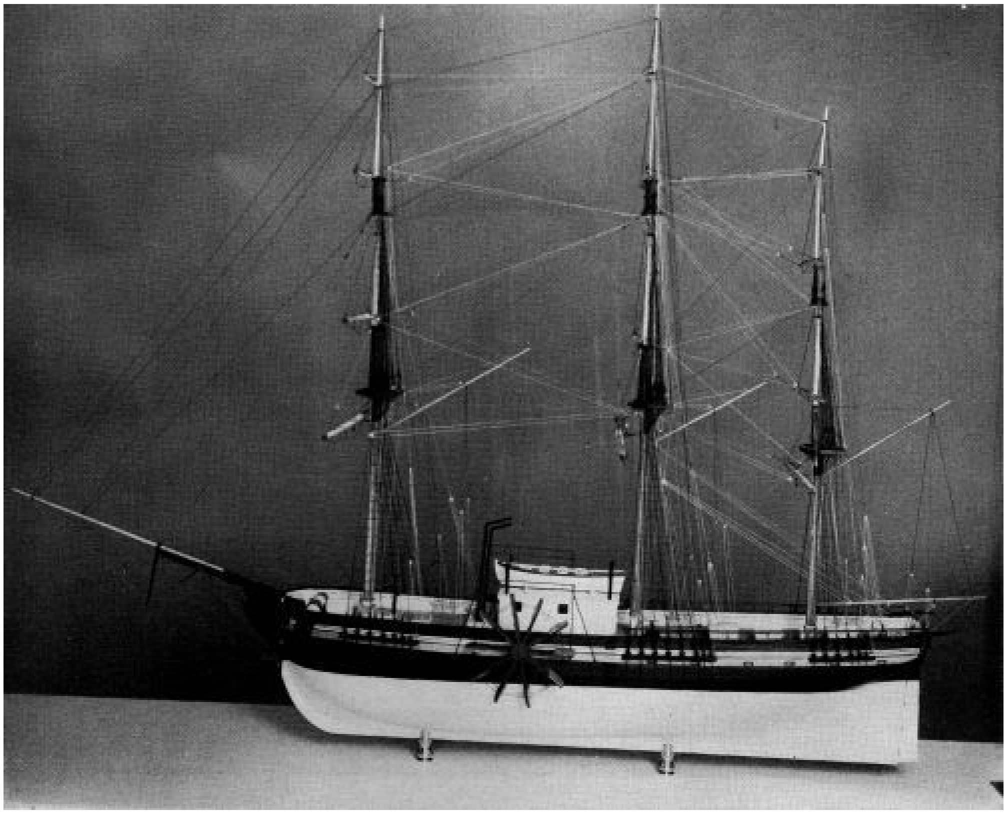

Figure 2.—The United States National Museum's new model of the *Savannah*. This model was built by Arthur Henning, Inc., of New York City, from the ship's plans as reconstructed by staff members of the Museum's division of transportation. (*USNM* 319026.)

The first step in the research for creating a more faithful representation of the *Savannah* was to obtain the customhouse description of the ship. It was readily established that she was built as a sailing packet ship by the Fickett and Crockett shipyard[4] at Corlaer's Hook, East River, New York, and that she was launched August 22, 1818. Her register shows that she was 98 feet 6 inches in length between perpendiculars, 25 feet 10 inches in beam, 14 feet 2 inches depth in hold, of $319^{70}/_{94}$ tons burthen, and with square stern, round tuck, no quarter galleries, and a man's bust figurehead.

These dimensions of the *Savannah* required the researchers to investigate the method of taking register dimensions in 1818. It was found that the customhouse rule then in effect measured length between perpendiculars above the upper deck, from "foreside of the main stem" to the "after side of the sternpost." The beam was measured outside of plank at the widest point in the hull, above the main wales. If a vessel were single-decked, the depth was measured alongside the keelson at main hatch from ceiling to underside of deck plank; if double-decked, one-half the measured beam was the register depth.[5] However, inspection of the register of a number of ships of 1815–1840 showed that, in practice, double-decked ships commonly were measured as single-decked ships; this obviously was the case in the *Savannah*. Also, due to the lack of precise measuring devices, the register dimensions were not always accurate, particularly those of the length, which often were in error as much as one foot in a hundred, as was found by investigation of various classes of vessels. Because of inherent difficulties in measuring to the required points, this condition lasted even after steel tapes were introduced late in the 19th century.

The Museum's researchers next turned their attention to examination of the Marestier work, a French report on early American steam vessels that had become known to some American marine historians in the 1920's. The author was a French naval constructor who, on orders from his government, had spent two years in the United States between 1819 and 1822 studying American steam vessels, schooners, and naval vessels. The published report contained only material on steam vessels and schooners. The portion dealing with naval vessels was not published, and the manuscript has not been found to the present time (1960). The publication, a rare book, was available in only a few collectors' libraries or public institutions in the United States. In 1930 the writer translated the chapter on schooners,[6] and in 1957 Sidney Withington translated most of the remainder.[7] As a result of these publications and earlier published references, the Marestier material became widely known to persons interested in ships.

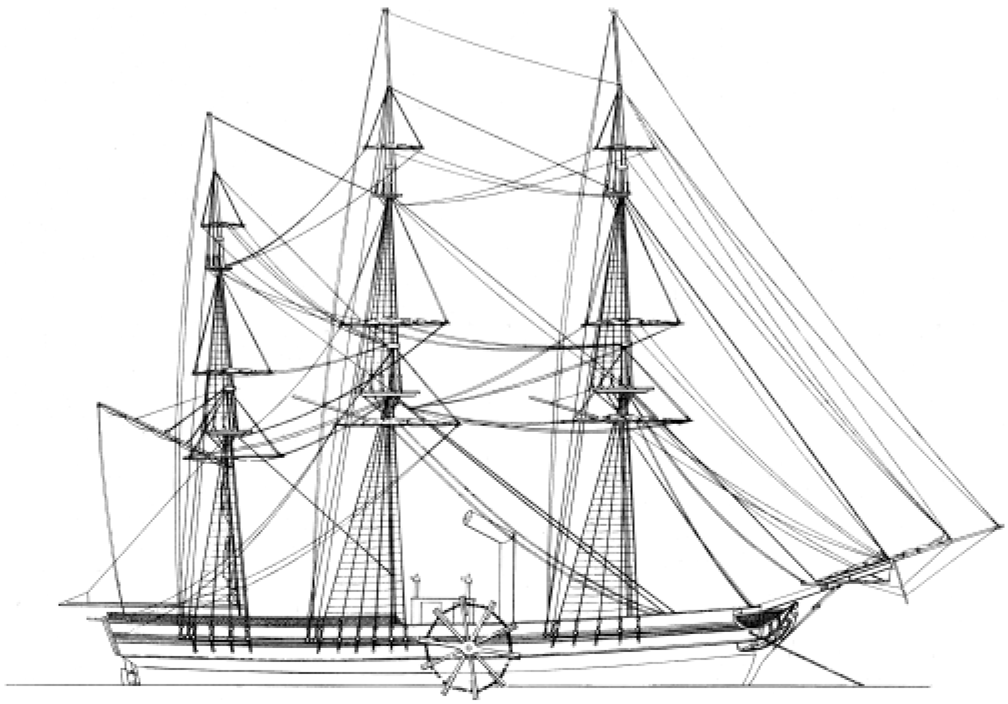

Figure 3.—Marestier's sketch of the *Savannah* (from plate 8 in Withington's translation of the Marestier report). Heights of lower masts are excessive by all known American masting rules; and, according to Marestier's drawing of the engine (see figure 4), the deckhouse is too short.

Withington's translation states that the *Savannah* measured 30.48 meters (100 feet) in length and 7.92 meters (26 feet) in beam and that she drew 3.66 meters (12 feet) in port and 4.27 meters (14 feet) loaded. Marestier's sketch (see fig. 3) of the outboard of the *Savannah* shows a ship-rigged, flush-decked vessel with a small deckhouse forward of the mainmast and nearly abreast of the side paddle wheels. The stack is a little forward of the deckhouse and has an elbow at its top. Netting quarter-deck rail is shown and a bust figurehead is indicated. The position of the hawse pipe shown at the bow indicates the wheel shaft to have been at or about deck level. For structural reasons, and in compliance with the sketch, the wheel shaft would have been just above the deck.

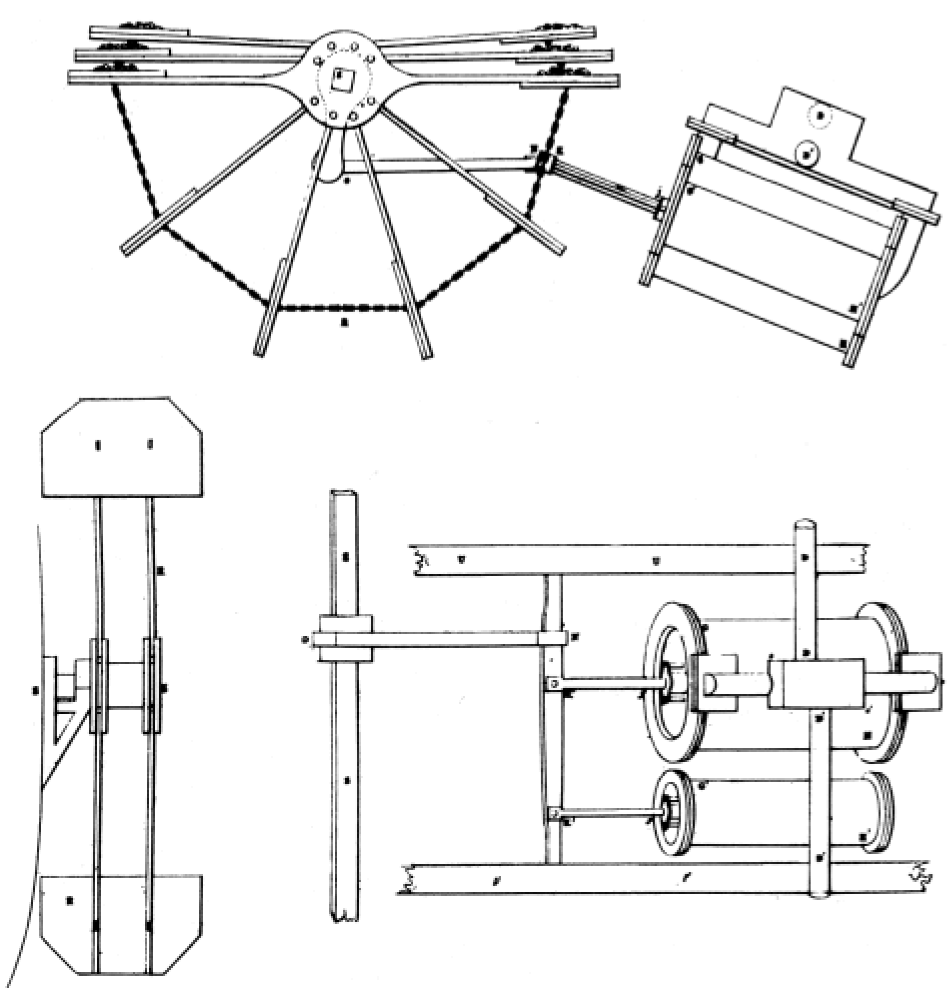

Figure 4.—Marestier's drawings of the *Savannah's* engine (from plate 7 in Withington's translation of the Marestier report). The graphic dimensions do not precisely correspond to the scale of dimensions in Marestier's text, nor with other recorded measurements.

Marestier's drawings of the engine and paddle wheels[8] are reproduced in figure 4. The nonoscillating engine is inclined toward the paddle-wheel shaft. The connecting rod operates a crosshead to which is pivoted a pitman, or oscillating rod, that operates the paddle-wheel crankshaft. Alongside the steam cylinder is an air pump cylinder, also connected to the crosshead. The steam inlet and outlet pipes enter a valve chest on top of the steam cylinder, which is described as being 1.035 meters (3.4 feet) in diameter, and of 1.5 meters (4.9 feet) in stroke.

The paddle wheels are shown as being of iron, with two fixed arms opposite one another on the hub. The other arms (four above and four below the fixed arms) are pivoted to the hub and held spread by chain stays. These eight blades fold, in pairs, to each of the fixed arms. The wheels are shown in elevation, with the upper pivoted arms folded on top of the fixed arms, and in cross section; the latter shows the shape of the buckets, hub, and outboard bearing of the shaft. The wheels are described as being 4.9 meters (16 feet) in diameter, while the buckets are 1.42 meters (4.65 feet) wide and 0.83 meters (2.72 feet) deep. The two outer corners of each bucket are snyed off at nearly 45°. The wheels are shown folded in the sketch; according to the description, they could be unshipped from the shaft and stowed on deck when desired. The method of removing the wheels from the shaft is not described, but from the drawings it seems probable that they were detached from the shaft by removing a lock bolt outboard and sliding the wheels off the square shaft. The hub seems adequate for this. Marestier states that this removal could be accomplished in 15 to 20 minutes; the logbook shows that it took 20 to 30 minutes to perform this operation at sea.

Marestier states that the ship had spencer masts and trysails on fore and main, and a spencer mast on the mizzen for a spanker; he illustrates these as having royal poles, but with no royal yards crossed.[9] The smokestack is described as pivoted. The mainstay is double, setting up at deck, near rail, and forward of the foremost shrouds of the foremast to clear the stack and foremast.

The boilers were in the hold, but Marestier gives no dimensions. However, he comments that, in American steamers, the space for steam in the boilers varied from 6 to 12 times the capacity of the cylinder. He gives the *Savannah's* boiler pressure as 2 to 5 pounds per square inch and the maximum revolution of the wheels as 16 revolutions per minute. The boilers could burn coal or wood. Judging by Marestier's sketch of the ship, the stack was at the firebox end; the boiler or boilers were underneath the engine.

The log of the *Savannah* gives little useful technical information other than that the ship readily made 9 to 10 knots under sail in fresh winds, showing she could sail well. Under steam alone the log credits the ship with a speed of 6 knots; Marestier estimated her speed at $5^1/_4$ knots in smooth water. The log shows that she usually furled her sails when steaming, though on a few occasions she used both steam and sail. In her crossing from Savannah to

Liverpool she appears to have been under steam for a little less than 90 hours in a period of about 18 days (out of the total of 29 days and 11 hours required to cross). There is no evidence of any intent to make the whole passage under steam alone, for the vessel was intended to be an auxiliary, with sails the chief propulsion.

Captain Collins states in his notes that the ship was built by Francis Fickett as a Havre packet, that she stowed 75 tons of coal and 25 cords of wood, and cost $50,000. Apparently quoting Preble[10] to a great extent, he also states that the engine developed 90 horsepower and had a 40-inch diameter cylinder with a stroke of 5 feet.

Preble states that the ship was purchased for conversion to a steamer after launching and gives statements by Stevens Rogers, sailing master of the *Savannah*, to the effect that the ship was built as a Havre packet and that the project ruined financially one of the investors, William Scarborough. Rogers, who made these statements in 1856, also said the ship was built by "Crocker and Fickett." Contemporary newspapers, quoted by Preble, state that the ship had 32 berths in staterooms for passengers.

Morrison[11] credits the building of the *Savannah* to Francis Fickett and says she was intended for the Havre packet run. He states that the vessel cost $50,000; that her paddle wheels, each with eight buckets, were 16 feet in diameter; and that she had canvas wheel boxes supported by an iron frame. Morrison also relates the history of the ship after her return from Russia—the removal and the sale of her machinery to James P. Allaire, the operation of the ship as a sailing packet between New York and Savannah under the ownership and command of Captain Holdridge, and her stranding and loss during an east-northeast gale on November 5, 1821, at Great South Beach, off Bellport, on the south shore of Long Island. He also states that the steam cylinder of her engine was exhibited at the Crystal Palace Fair in New York during 1853, and that the ship proved uneconomical due to the large amount of space occupied by the engine, boilers, and fuel, leaving little space for cargo. Morrison apparently used some of the statements made in 1836 and 1856 by Stevens Rogers, who was the sailing master on the famous voyage.

Tyler[12] names the stockholders of the Savannah Steamship Company, owner of the *Savannah*. The company was proposed by Capt. Moses Rogers, and its shareholders were William Scarborough, John McKenna, Samuel Howard, Charles Howard, Robert Isaacs, S. C. Dunning, A. B. Fannin, John Haslett, A. S. Bullock, James Bullock, John Bogue, Andrew Low, Col. J. P. Henry, J. Minis, John Sparkman, Robert Mitchell, R. Habersham, J. Habersham, Gideon Pott, W. S. Gillet, and Samuel Yates. Tyler establishes, by the company's charter, that the objective was to institute a New York-Savannah packet service, for which the *Savannah* was to be the first ship. He shows that, due to the economic depression of 1819, the *Savannah* sailed to Liverpool in ballast and without passengers. Her fuel capacity is given as 1,500 bushels (75 tons) of coal and 25 cords of wood. [It should be noted that 1,500 bushels of bituminous coal does not quite equal 75 tons.] Tyler quotes S. C. Gilfillan[13] as to criticisms of the engine and its design.

Partington[14] estimated coal consumption to be nearly 10 tons a day; remarked on the uneconomical arrangement of the ship, with the engine and boiler occupying the greater part of the space amidships, between fore and main masts; and located the axle of the paddle wheel "above the bends," that is, in the topsides above the wale. The description he gives of the unshipping of the wheels is that the pivoted blades were removed and the fixed blades, in horizontal position, were left on the shaft. This agrees with a Russian description referred to later. The logbook repeatedly speaks of "shipping" and "unshipping" the paddle wheels, indicating that the wheels were entirely removed from the shafts and stowed on deck.

Watkins[15] showed, by the account books of Stephen Vail, owner of the Speedwell Ironworks near Morristown, New Jersey, that the engine was built by Vail, but apparently to designs by Daniel Dod. The latter built the *Savannah's* boiler at Elizabeth, New Jersey, and made some parts of the engine, which he furnished, incomplete in some instances, to Vail. These account books, which were in the possession of John Lidgerwood of New York City in 1890, show the steam cylinder to have had an inside diameter of $40^3/_8$ inches and a 5-foot stroke. Reference in the account books to an error in Dod's draught of a piston proves that Dod designed the engine.

Watkins states that the engine was rated at 90 horsepower. He does not give the diameter of the pump cylinder, but, judging by the scaling of Marestier's drawing and by a rather indefinite entry in the Vail account book, it appears to have been between 17 and 18 inches. Quoting Captain Collins at some length, Watkins writes that the mainmast was placed farther aft than was usual in a sailing ship, and that the vessel had a round stern. Collins apparently based his opinion upon an unidentified "contemporaneous lithograph" and upon "all other illustrations of this famous vessel." Collins' conception of the appearance of the *Savannah* is shown in a drawing by C. B. Hudson that is reproduced as the frontispiece in Watkins' publication. A statement by Stevens Rogers that was published in the *New London Gazette* in 1836 appears to have been the original source for statements regarding the *Savannah's* fuel capacity, her sale, and her loss in 1821 while owned and commanded by Capt. Nathaniel Holdridge, "now master of the Liverpool packet ship *United States*." Watkins also gives a picture of Stevens Rogers' tombstone, on which there is a small carving purported to be of the *Savannah*. The tombstone was made in 1868.

From a Russian newspaper contemporary with the *Savannah's* visit to St. Petersburg, Frank Braynard found a statement that the vessel had two boilers, each 27 feet long and 6 feet in diameter.[16] It was also shown she had at least one chain cable. Considerable information on the cabin arrangement and the method of folding the wheels was also obtained from this Russian source.

In spite of a very extensive bibliography on the *Savannah*, the basic sources for reliable technical description are Marestier's report on American steamers, the logbook of the ship, Watkins' extracts from the Speedwell Iron Works account book, the customhouse records, and some of the statements made by Stevens Rogers between 1836 and 1856. Plans of the ship, or a builder's half-model, have not been found. Marestier's sketch of the *Savannah*, which is not a scale drawing, and his drawings of the engine and paddle wheels were the only available illustrations upon which reconstruction could be based.

Through the efforts of Malcolm Bell, Jr., of Savannah, Georgia, and Frank Braynard, a search was made by Russian authorities at Leningrad for contemporary references to the ship. This work resulted in information as to how the side wheels were folded, the dimensions of the boilers, and some description of the cabins and fittings.

As to the ship itself, the customhouse registered dimensions are of prime importance; they fix the over-all hull dimensions within reasonable limits. A vessel of 1818 measuring 98 feet 6 inches between perpendiculars would have been 100 to 104 feet long at rail. The type of ship represented by the *Savannah* is well established. All references are in agreement that she was built as a packet ship—a Havre or transatlantic packet in most accounts.

The packet ships listed by Albion[17] show that all the pioneer ships of the transatlantic Black Ball Line—which began operation with the sailing of the 424-ton *James Monroe* on January 5, 1818—measured at least 103 feet 6 inches between perpendiculars. Two of the pioneer ships of the first Havre Line—which did not begin operation until 1822—were under 98 feet between perpendiculars. The second Havre Line began operation in 1823; of its four pioneer packets, two were purchased general traders measuring under 98 feet between perpendiculars. The coastal packets built between 1817 and 1823 were all under 100 feet between perpendiculars. It is apparent, then, that the size of the early packets did not indicate, with any degree of certainty, the trade in which they might be employed.

Belief that the *Savannah* was built as a Havre packet is based upon Stevens Rogers' statements, and her size obviously does not make this impossible; nevertheless, it seems highly improbable that she was built for the Havre service because no Havre line of packets had been organized as early as 1818 out of New York or Savannah so far as can be found. However, the matter is not of very great concern as it is probably true that the models of coastal and transatlantic packet ships were quite similar at the period of the *Savannah*. This statement is supported by the plan of a coastal packet built seven years after the *Savannah*.

The hull-type of these early packets can be established. While no half-models or plans of packets built before 1832 could be found, offset tables of a Philadelphia-New Orleans packet of 1824–1825 were obtained through the courtesy of William Salisbury, an English marine historian who had been studying the British mail packets. These offset tables had been sent from Washington on March 25, 1831, by John Lenthall, U.S. naval constructor, to William Morgan and Augustin Creuze, London editors, for publication.[18] The offset tables were for a packet ship 103 feet between the perpendiculars of the builder (rather than between those of the customhouse) and 27 feet moulded beam. An examination of the files on American packet vessels in the collection of Carl C. Cutler, curator emeritus of the Mystic Marine Museum, showed with certainty that the offsets were for the *Ohio*, built at Philadelphia late in 1825. The drawings of this ship (fig. 5) were made from the offset tables and from other measurements; minor details are from portraits of packet ships, particularly of the first *New York* (1822–1834) of the Black Ball Line.

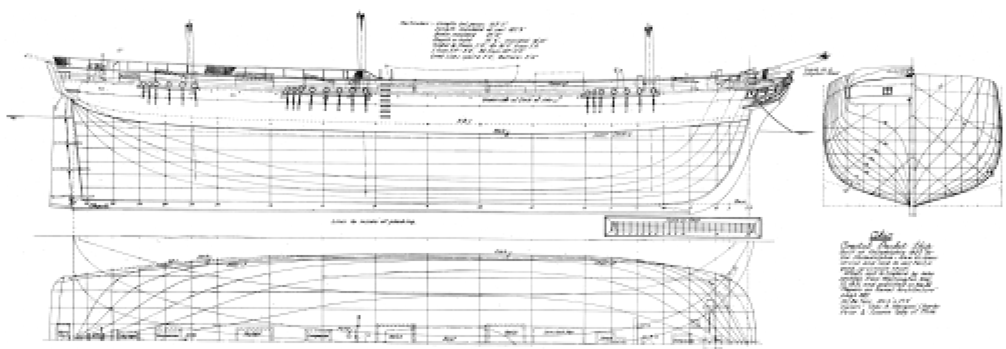

Figure 5.—Lines of the coastal packet ship *Ohio*, built at Philadelphia in 1825 for the Philadelphia-New Orleans run. The *Ohio* represents the general type of early American packet ships.

VIEW FULL SIZE

The *Ohio* was two-decked, with the upper deck flush. She had rather straight sheer, 27-inch bulwarks, a moderately full but easy entrance, a fine, long run, and little drag to the keel. The midsection was formed with moderately short and rising floor, round and easy bilge, and some tumblehome in the topside. The stem raked a good deal for a ship-rigged vessel; the post raked slightly. There was a distance of 6 feet between upper and lower deck planks. The stern was of the square transom, round tuck form, as mentioned in the *Savannah's* register. Lenthall reported the *Ohio* to have been a good sailer and to have had other desirable qualities. She was registered as being of 351.86 tons burthen, 105.5 feet between perpendiculars, and 27.4 feet in extreme beam. She was, therefore, about 7 feet longer and about 2 feet 3 inches wider than the *Savannah*. The plan shows she was about 2 feet 4 inches deeper in hold than the *Savannah*, and, according to Cutler, she had "an unexpected degree of sophistication for a coastal packet of that period."[19] By modern standards, the *Ohio* shows a well-advanced design for the period.

Reconstructing the Plans

The first step in the reconstruction of the *Savannah's* plans was to block out the register dimensions on a scale of one-quarter inch to the foot in a drawing and then to work out the profile, using the *Ohio* plan as a general guide. This produced a hull about 100 feet 9 inches in length at main rail to inside of plank, or "moulded"; 25 feet 6 inches moulded beam, allowing 3 inches for plank (as usual in a ship of this size and date); and about 15 feet 4 inches moulded depth at side, keel rabbet to underside of upper deck. The bulwarks were drawn at 28 inches height. Next, the mast positions were decided by prorating from the plan of the *Ohio* the position of each mast from the fore perpendicular and then modifying these positions slightly by use of masting rules contained in M'Kay's book[20] of 1839.

Since it appears that the *Savannah* may not have been purchased for conversion to a steamer until near the date of her launch and because of the lack of identification of the lithograph referred to by Collins, the statement that the mainmast was placed farther aft than normal was rejected. At launch her mast partners would have been in place and the deck laid. Any alterations in the position of the mainmast then would have made it impractical for the owners to demand them of the builders without heavy additional expense. In addition, the plan, as it was developed, indicated no need for such alteration.

The plan of the engine, drawn to the same scale as the profile plan, was shifted about on the lower deck in the hull profile to determine where the engine and side paddle wheel shaft might be located. A little experimentation and study made it certain that the proper location could be estimated within a foot or so, to scale, as to fore and aft positions. The after end of the cylinder, and its piping, had to clear the mainmast by at least 9 to 10 inches to allow removal of the cylinder head for inspection and repair. The position of the wheels, stack, and masts in Marestier's sketch of the ship make it certain that the engine was on the lower deck, abaft the paddle wheel shaft. Due to differences between the dimensions stated by Marestier and in the Vail account books and what the graphic scale in Marestier's engine drawings produce, the exact dimensions of the engine are uncertain. Nevertheless, they can be approximated with enough accuracy for our purpose. As a result of this treatment, it seems fully apparent that the engine was abaft the paddle wheel shaft, with frame extending abaft the mainmast on the lower deck; there does not appear to be a practical alternative in the light of the available evidence. This matter will be referred to again.

The size of the cylinder and its valve chest and the inclined position of the cylinder indicate conclusively that the valve chest was in the mainhatch, which would normally be just forward of the mainmast. Even then, the after flange of the cylinder would just clear the lower deck, allowing 6 feet between decks, as in the *Ohio*. The cylinder would have been about 6 feet long; the graphic scale indicated 6 feet 3 inches. The diameter of the cylinder plus height of valve chest seems to have been 5 feet 9 inches to 6 feet. Because of the use of the crosshead and a connecting rod, pivoted at crosshead, the oscillating rod (or pitman) and piston together equalled twice the stroke plus allowance for stuffing box, crosshead, and pitman bearings. Therefore, the engine's over-all length, from head of cylinder to the centerline of the side paddle wheel shaft, could not have been much less than 15 feet 9 inches, and probably as much as 16 feet 2 inches, thus making the length at extreme clearance of crank throw as much as 19 feet. These dimensions indicate that the centerline of the side paddle wheel shaft must have been from 38 to 39 feet from the forward perpendicular. It is not clear how the wheel shaft was mounted in the vessel. Taking into consideration her depth and her reported draught, light and loaded, the Marestier sketch, and the hull structure then used, it seems reasonable to place the centerline of the shaft (which seems to have been about 7 to 8 inches square) about 12 inches above the upper (or spar) deck to allow proper dip of the blades. This position would have given proper blade immersion at the mean draught of 13 feet.

In order to get the engine below deck, and to get the boiler or boilers placed, it was necessary to cut a large opening in the two decks. It may be assumed that this opening was big enough to take the cylinder, without valve chest, and also the boilers, which went into the hold. Taking the proportions of other boilers as shown by Marestier, it has been estimated that the *Savannah* might have had a boiler about 18 to 20 feet in length, 7 to 8 feet wide, and 6 to $6^1/_2$ feet high at firebox. The form might be the same as that of *Fulton the First*, illustrated in the translation of Marestier's report.[21] However, since the Russian descriptions[22] indicate there were two boilers, each measuring 6 feet in diameter and 27 feet in length, the two boilers would have reached past the mainmast if they were located in the same manner and in the same place as the boilers shown in the illustration of *Fulton the First*. Consequently, if the Russian description is accepted, there would have been a need for longer fuel (coal) spaces in the wings.

The boilers, then, were the largest piece of equipment to be passed through the decks; for this an opening (estimated to have been about $10^1/_2$ feet wide and $8^1/_2$ feet long) probably was cut through both decks about 3 feet forward of the main hatch, which was commonly a little forward of the mainmast. The boilers could then have been lowered, after end first, into the hold. The opening in the lower deck could then have been closed, except for a small hatchway perhaps, and the steam cylinder let down to the lower deck and moved aft into position. To allow the crosshead to reach its maximum travel, the opening in the upper deck would have been about $10^1/_2$ feet wide—the over-all width of the engine frame—and would have been left open, inside the deckhouse.

The width of the boilers might be particularly important because it would determine the deadrise at floor in the hull. The apparently precise dimensions of the boilers given in the Russian description were utilized to arrive at a suitable hull form. Both a single boiler and a double boiler (as described in the Russian accounts) were placed in the hull to assure the correct space estimates.

Since the engine, as shown by Marestier, had an air-pump cylinder alongside the steam cylinder (with the pistons of both attached to the crosshead), it is evident that a condenser was employed. This condenser would not have been much larger than the air-pump cylinder. It may have been placed under the side paddle wheel axle on the lower deck, but its mode of operation is unknown. Possibly it was of the jet type, with pumps operating off the paddle wheel axle and with a return of condensate from a hot well into the feed water line. A number of possibilities could be mentioned, all speculative. However, there was no doubt that this equipment could be properly installed in the reconstructed hull, either on the lower deck or in the hold.

Two questions have been raised as to machinery arrangement—whether the engine, and boilers also, might have been forward of the wheel shaft, and whether the wheel shaft was above or below deck. If the engine were placed forward of the wheel shaft, the wheels might be farther aft than is proposed in the reconstruction. However, the smokestack could not then be forward of the wheel shaft as shown by Marestier because it would have had to pass through the engine frame, thus interfering with the movement of the large crosshead. If the engine were abaft the wheel shaft, the stack could have been only as shown by Marestier. The boilers might then have been forward of the wheel shaft only if the stack were at the end away from the firebox. However, the length of the boilers as indicated by the Russian description would then have required them to pass through the bows!

Models have been built of the *Savannah* in which the engine and boilers are forward of the paddle wheel shaft, and the shaft below the main deck. This was accomplished by placing the engine off center so that the stack came through the decks alongside it. This is an impractical arrangement because it would have created an impossible ballasting problem. The weight of the engine, to port in the models, would have to have been counteracted by ballast to starboard. Due to the coal bunkers, and the possibility of two boilers below the engine in the hold, there would not have been room for sufficient ballast. In addition, were such ballasting possible, the combined weights were too far forward to give proper trim, and a great deal more ballast would have been required far aft, a most impractical proceeding.

The position of the wheel shaft was determined as described earlier. The ship was apparently well-advanced in construction at the time of purchase. Her clamps and shelves supporting her upper deck beams, which then would have been in place, were important strength members. In reconstructing, to place the wheel shaft below these members would not only bring the engine nearly level—it is described and shown inclined by

Marestier—but also would immerse the paddle blades too deeply for the draft and depth of the hull. To place the shaft below or through the lowest clamp member would require the shaft centerline to be at least 3 feet below the upper deck, and this would contradict Marestier. These questions indicate the importance of a scaled drawing when deciding arrangement in the reconstruction of a ship under the circumstances existing in the *Savannah*. Some models have been built with the shaft below deck by disregarding the structural and dimensional objections just outlined.

The question of the number of boilers originally was raised by Braynard. A single boiler with double flues was a common boiler design in American steamboats of 1818–1828, and this form of boiler is shown in a number of Marestier's drawings. In general descriptions, "boiler" and "boilers" are often used interchangeably, and this probably came about through confusion over the number of flues. A "single boiler, double flues," would thus become "boilers," apparently. The Russian description specifically states there were two boilers, and gives specific dimensions; though these probably are not exact. Either a single boiler with double flues, or double boilers, each with a single flue, could have been fitted in the reconstruction. However, fuel space is affected and, with double boilers, the cross-sections of the bunkers are reduced to about 20 square feet each; therefore, the bunkers would have to become much longer. It may be said that the boiler capacities in relation to dimensions of the steam cylinder as indicated in the Russian description far exceed those given by Marestier. As a practical matter of ship design, it seems that the single boiler would have been a more logical fitting than double boilers. The boilers were apparently of copper, and expensive. However, this matter does not affect the hull-form and dimensions established for the reconstruction, as the drawings proved. The Russian description does show that the cargo space was extremely small and practically nonexistent, indicating the effect of the large boiler capacity.

All requirements that have been given can be approximated for space necessary in the hull. It is established that the ship carried about 75 tons of coal and 25 cords of wood. The coal would take up from about 1,700 to 1,850 cubic feet of space, and because of its weight it would have to be bunkered alongside the boilers in the lower hold, where there would be ample room, in the reconstruction, for two bunkers, each in excess of 30 square feet in cross section and about 28 feet in length for a single boiler; one third more bunker space, in length, would be required for double boilers. Such bunkers would together hold about the required tonnage or cubic footage. The cord wood would have required, say, two bunkers each of about 60 square feet in cross section and 20 to 24 feet in length. Because of the light weight, the cord wood could have been stowed in the wings on the lower deck. There is room for the required stowage on the lower deck in the reconstructed hull, leaving ample passages under either side of the engine frame.

Marestier shows the location of the stack as being abreast the buckets on the forward side of the paddle wheels, and it has been so placed in the reconstruction. The deckhouse shown in Marestier's sketch extends from a little forward of the mainmast to a little forward of the paddle wheel axle. Probably this house actually covered the main hatch and the crank-connecting-rod hatchway; therefore, Marestier shows it too short. In the reconstruction, the deckhouse works out as between 17 and 18 feet long. Its width can only be guessed at, but it probably would have been as wide as the opening cut in the upper deck for machinery—say 11 feet. Perhaps this house contained the engineer's stateroom and that of his assistant, as well as a ladderway to the engine room. Doors on the sides of the house gave access to these spaces and to the inboard shaft bearings. Bunker hatches were probably forward of the house and outboard; these are taken as being about 2 feet 6 inches wide and 3 feet 6 inches long—large enough to allow coal baskets to be lowered through them, as well as to allow cord wood to be passed below.

A fidley hatch, in which the stack passed through the upper deck, would have been a square hatch forward of the deckhouse. This hatch, about $2\frac{1}{2}$ to 3 feet square, would have been fitted with an iron or iron-bound fidley grating, with solid cover over. The stack could have been swivelled, to bring the elbow to leeward. The upper portion of the stack probably overlapped the lower portion at least 3 to 4 feet above the fidley coaming, and the upper stack rested on a collar bearing at the bottom of the overlap. Perhaps straps were bolted to the side of the upper stack to take heaving bars athwartships, by which two men could rotate the upper stack to turn the elbow to leeward.

The bearings of the paddle wheel axle were perhaps four in number. Two, one either side of the crank, may have been secured to the engine frame just inside the deckhouse walls. Two were certainly outboard, one on each side, fastened to the topsides, as shown in Marestier's sketch of the wheel construction. The axle, probably square in cross section, turned only at the bearings and wrist pin. It may have been cast in two parts, each with a crank arm, and then joined by the wrist pin, after the latter had been turned.

The wheels, shown in much detail in Marestier's sketches of the engine, had flanged hubs to which the pivoted arms or spokes were bolted. The fixed arms were integral parts of the outer hubs. The inner flanges were cast with the hubs. To fold the blades, the fixed arms were brought parallel to the rail, then the chain span between each pair of the pivoted blades on top of the wheel was disconnected and a pair of the blades, each way, were dropped on top of the fixed arms, or blades, and lashed there. The wheel was then given a half-revolution and the process repeated. The wheel could then have been unshipped from the hub by sliding it off the square shaft end after removing, let us suppose, a bolt or pin in the hub. Some writers, like Collins, refer to a "jointed" or "hinged" axle, but Marestier makes no mention of such an arrangement; indeed, his sketch makes a "broken" axle impractical. The wheels could have been removed from the axle and lifted aboard by use of tackles from the main yard ends, or from a fore spencer gaff if it were made long enough. However, as stated in the Russian description, the pivoted blades were removed and stowed aboard, leaving only the two fixed arms in a horizontal position outboard. This is a far more convenient treatment than unshipping the whole wheel, as might be supposed from logbook mention of "shipping" or "unshipping" the wheels.

There remain some other matters to be explored. The ship was fitted with 32 passenger berths in staterooms. The passenger accommodations for first class passengers in the early (1820–1830) packets were aft, on the lower deck. The berths would have been about 6 feet 2 inches long, and $2^1/_2$ feet wide. With berths placed athwartships and allowing for cabin bulkheads, there would have remained a space at least 10 to 12 feet wide down the centerline of the ship. This space would have provided space for a mess table and a lounge area. Each stateroom would then have been about 7 feet long fore and aft and could have contained four athwartship berths. The space available abaft the middle of the after cargo hatch would have allowed four staterooms on each side and room at the extreme stern for a small master's cabin, with toilets on each side. The cabin of the mates and stewards, containing two berths each, would then have been about abreast of the fore end of the after cargo hatch.

The galley would have been on the lower deck, just abaft the foremast and forward of the fore cargo hatch. Food would have been carried aft along the lower deck to the cabin, by way of passages on either side of the engine frame. Cabin stores would have been in the hold below the passenger accommodation, and here food, water, and other stores would have been kept. A small cargo space, say of about 1,500 to 2,500 cubic feet, depending on bunkers, would have been possible in the after hold. A fore cargo hold of about 1,000 to 1,500 cubic feet of contents could be expected; forward of this would have been sail locker, spare rigging gear, and a cable tier. On the lower deck, above these spaces, a forecastle might have had berths for 12 to 14 men. The cables and chain would be passed through the forecastle to the cable tier below by chutes leading from cable scuttles in the upper deck abaft the windlass on each side of the centerline of the ship.

The upper deck, abaft the mainmast, was reserved for use of the passengers and officers of a packet. The low, 28-inch bulwarks were insufficient to give proper protection there, so they were increased by employing a 16-inch rail made of a cap supported by iron stanchions above the main rail. This rail was closed in by a tarred netting extending from the main rail upward to the quarter-deck rail cap and running from the mainmast aft to the stern. This is plainly shown in Marestier's sketch of the *Savannah* as well as in some portraits of early packet ships.

Though the passenger accommodations described were far from palatial by modern standards, they were considered adequate in the 1820's and for almost 15 years afterwards. The staterooms had no individual toilets. Usually there were two small toilets, one on each side of the stern cabin, at the extreme stern on the lower deck, in the quarters. Usually the master's stateroom and toilet were to starboard, with a public space and toilet to port. Sometimes toilets for the crew were placed forward, on either bow abaft the catheads on the upper deck. These were small cabinets accommodating one person each, and with the door closed for privacy there was not room to stand. To enter the user backed in, crouching. Such cabinets are not shown by Marestier, so probably the crew used the headrails, as then was usual in merchant vessels.

The hull-form to be chosen had to enclose all spaces that have been described or listed. Since the *Savannah* is known to have sailed quite fast for her length, her lines had to equal those of the *Ohio*; however, her smaller size and other factors indicated a somewhat different hull-form, with harder turn of the bilge and a little less deadrise. Due to the position of the machinery, the effect of its weight and that of the necessary fuel had to be considered. The midsection, or cross section of greatest area, would have to have been only a little abaft the paddle wheel axle to allow proper trim with a minimum of ballast. It was found by this criterion that the midsection of the reconstructed hull was located in proportion to length in a comparable manner to that of the *Ohio*. The run could have been made about as long and easy, in proportion, as that of the *Ohio*; likewise, the entrance could have been equally well designed for sailing. Probably a little ballast—stone, gravel, sand or pig iron—was required under the temporary flooring of the cargo holds, most of it abaft the mainmast. Some ballast would normally have been placed under the cabin stores, in the run. The boilers, engine, and fuel weights were relatively important. To trim the ship, with minimum ballast, the location of the machinery weights would have to have been about as shown in the reconstruction drawings. It may be observed that the engine and fuel weights are relatively great for the recorded hull dimensions and resultant displacement limitation, indicating only a small quantity of ballast would have been employed under any circumstance.

Using the *Ohio* as a guide, the midsection was formed to comply with the dimensions of the boilers and with due regard to the small dimensions of the *Savannah*. The result was a section having very moderate rise of straight floor, carried farther out in proportion to beam than in the *Ohio*, but with rather easy turn of the bilge and moderate tumble-home in the upper topsides. This section has a form found in plans of some American freighting ships of 1815–1830, but with slightly slacker bilge.

The stern used in the reconstruction was the "square stern and round tuck" seen in the *Ohio* and referred to in the *Savannah's* register. Collins' "round stern," shown in Hudson's drawing, did not come into use in America until about 1824, and then in naval ships only, so far as existing plans of American vessels show.

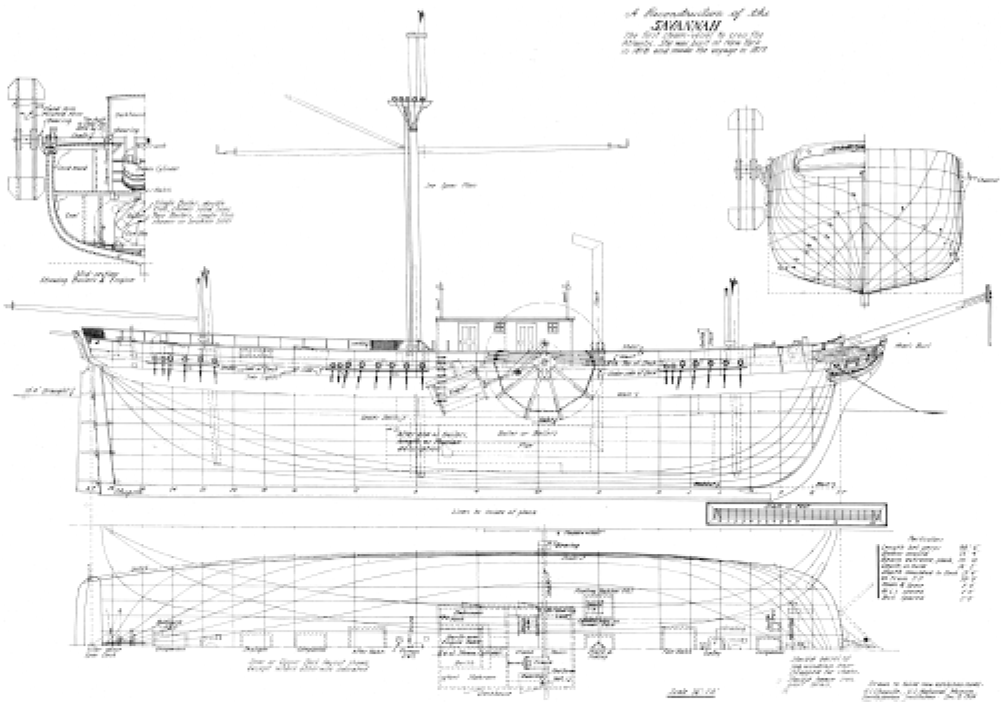

Figure 6.—Reconstruction of the hull lines and general arrangement of the *Savannah*.
VIEW FULL SIZE

The reconstructed hull-form (figure 6) shows the man's bust figurehead mentioned in the register, and the supporting head and trail mouldings employed in the packets and other American ships of the period. The figurehead may have had some relation to the original or intended name of the ship prior to her purchase for conversion. No detailed description has been found. A ship built to the drawing would at least sail well and would carry her machinery, fuel, etc., as indicated in the descriptions that exist. Whether or not the hull is precisely like that of the original ship can never be determined until the original plan, or model, is found. The proposed deck arrangement is shown in dotted lines, in plan view.

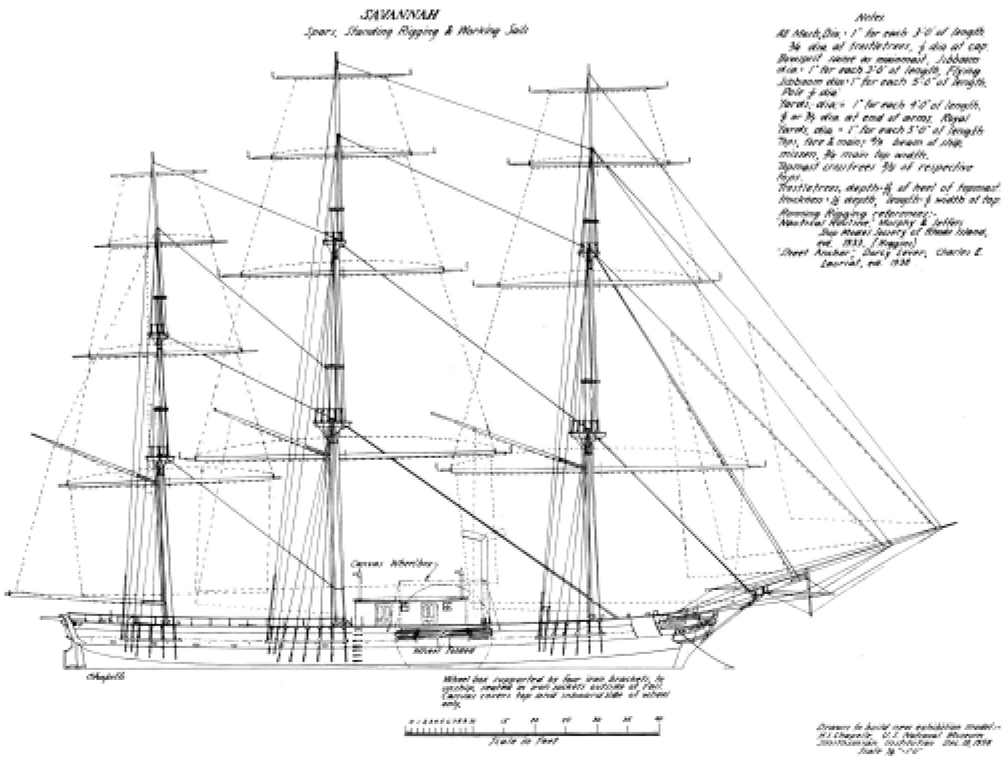

Figure 7.—Reconstructed drawing of spar and outboard profile of the *Savannah*. Dotted lines indicate working sails. Standing rigging only is shown. Royal yards were set flying and were crossed only when the ship was under full sail, never at anchor.
VIEW FULL SIZE

The rig shown in figure 7 is based upon Marestier's sketch and his incomplete description. Since the ship had long royal poles on her topgallant masts it is highly probable she crossed royal yards, like the later packet ships. The proportions for the length of spars are based upon the masting rules given by M'Kay[23] in 1839. The fore spencer gaff, used as a crane for handling coal and cargo if the fore or main yards were not available, may have been long enough to be used also as a crane to handle the side wheels. The stack and mainstays may have made the fore spencer sail a nuisance, so it may not have been set while the vessel had her engine. In general, aside from the use of the spencers on fore and main, the sail plan shown is of standard proportions and arrangement of 1815–1825. For rigging, Darcy Lever's book[24] was consulted. The drawing of the reconstructed *Savannah's* sail plan agrees with contemporary sail plans of ships in the author's collection. The log shows she set studding sails and had all the light canvas of a ship of her type.

There remain a number of matters that do not directly concern the reconstruction project but which are of sufficient technical importance to warrant comment. Apparently the engine was mounted on a wooden frame consisting of two large oak timbers on each side, say about 10" × 10", one above the other, that probably supported iron saddles in which the two cylinders rested. Between each pair would have been the iron track, or channel, in which the ends of the crosshead travelled, along the axis of the engine in elevation. These frames measured about 9 feet 2 inches, outside to outside, and reached from the beams of the upper deck on either side of the crank hatchway to abaft the mainmast on the lower deck. It is probable that the fore and after ends of the frame were supported by stanchions stepped on the lower deck at the fore end and in the hold at the after end. The crosshead was of iron and probably had shoes at the ends to work in the tracks or channels in the frame. To help steady the crosshead, these shoes probably were a foot or more long, for the loading of the crosshead is spread out. The pitman to the paddle wheel shaft is to starboard of the centerline of the engine; the steam cylinder piston is slightly off center of the frame and crosshead; and the piston of the air cylinder is close to the port engine frame. The steam lines to the valves of the steam cylinder come in horizontally over the frames. As has been mentioned, the frame may also have supported the paddle wheel axle bearings at the crank.

This engine has been criticized by some writers (see Tyler's[25] résumé of Gilfillan's[26] comments), but the *Savannah* logbook shows it gave no trouble, and should be compared with the logs of *Sirius* and *Great Western* as summarized by Tyler. The relatively slow piston speed and small power put little strain on the moving parts. Tallow was probably used for lubrication, being introduced into the valve chest by pots on top of the casing, where radiated heat would melt the tallow. From the valve chest the melted tallow was carried into the cylinder, and from there probably passed into the jet condenser. No doubt the lubricant became a sludge that had to be removed from the condenser at least once every 48 hours. There is no real evidence that the engine and boilers suffered any great strains; the operating pressure of steam must have been low at all times. The boilers were probably of very low efficiency and made steam slowly. Fuel consumption was high, and, according to the logbook, the vessel ran out of coal when she reached the English coast; however, she had enough fuel left to steam up the Mersey to Liverpool, probably using wood. At the time she ran out of coal she had used her engine about 80 to 83 hours. While this indicates a fuel consumption of almost a ton per hour, it must be remembered that the intermittent operation of the engine required expenditure of fuel to raise steam in cold boilers over and over again. This was one of the weaknesses in the auxiliary steamship, particularly, as in the case of the *Savannah*, when the engine was used a number of times during a voyage without long periods of continuous operation. Also, there is doubt that the vessel carried as much as 75 tons of coal; she probably had no more than 55 to 60 tons aboard, if the figure of 1,500 bushels is correct. It is impossible to establish exact weight-cubic measurements with the available data.

Though the authorities quoted seem to agree that the *Savannah* could steam only $4^1/_2$ to $5^1/_4$ knots in smooth water, her logbook credits her with 6 knots under steam alone at sea. However, this is probably an approximation affected by current and sea rather than a truly logged speed.

Judging by references in the logbook, the *Savannah* carried one boat on the stern davits. The davits, shown in Marestier's sketch, would handle a boat of about 16 to 18 feet in length. At sea the boat was probably carried on top of the deckhouse. The vessel obtained a new boat during her European trip. It is probable that the lack of passengers is why a second boat, which could have been stowed on the deckhouse roof, was omitted.

There is no record of how the *Savannah* was painted, except that the logbook refers to her "bright" strake. Packets appear to have followed what once was a Philadelphia practice in having a varnished band along the topsides. Marestier's sketch indicates that there may have been four or five bands of color, beginning at or a little above deck and wide enough for the top band to be up about two-fifths the height of the bulwarks. The hull was commonly black. The bands were red, white, and blue and there was a "bright" strake, or alternate black and varnished bands. These bands were about 3 to 5 inches wide. Sometimes the "bright" band, as mentioned in the *Savannah* logbook, was along the topside just above and adjacent to the top of the wale, or belt of thick planking, or might be the uppermost strake of the wale. Perhaps the *Savannah* had a wide bright band above the wale and multicolored bands just above the deck. The headrails were painted black, with mouldings at top and bottom of rails and with knees picked out with very narrow bands of yellow, or "beading." The figurehead was then commonly painted in natural colors, to suit the form of head if a figure or a bust. The bowsprit and davits probably were black. Deck structures were probably white, the neck natural, with waterways and inside of bulwarks white, the stack black, and rail caps varnished.

In this period it was unusual to copper a wooden ship before launch, so it is doubtful that the *Savannah* was copper sheathed. Since her voyage occurred during a period of financial depression, it is probable that her bottom was "white" (tallow and verdigris).

The reconstruction described herein produced a plan for a model that complied to the fullest extent with all the known dimensions and descriptions of the *Savannah* that have yet been found. The result showed that the United States National Museum's old model could not be altered to agree with the known features of the *Savannah* and that a new model was therefore necessary. So that the new model would be comparable to other models of early American steamers, existing or intended, in the Watercraft Collection, it was constructed on the scale of one-quarter inch to the foot. The new model (figs. 2, 8, and 9) is now on exhibition at the Smithsonian Institution.

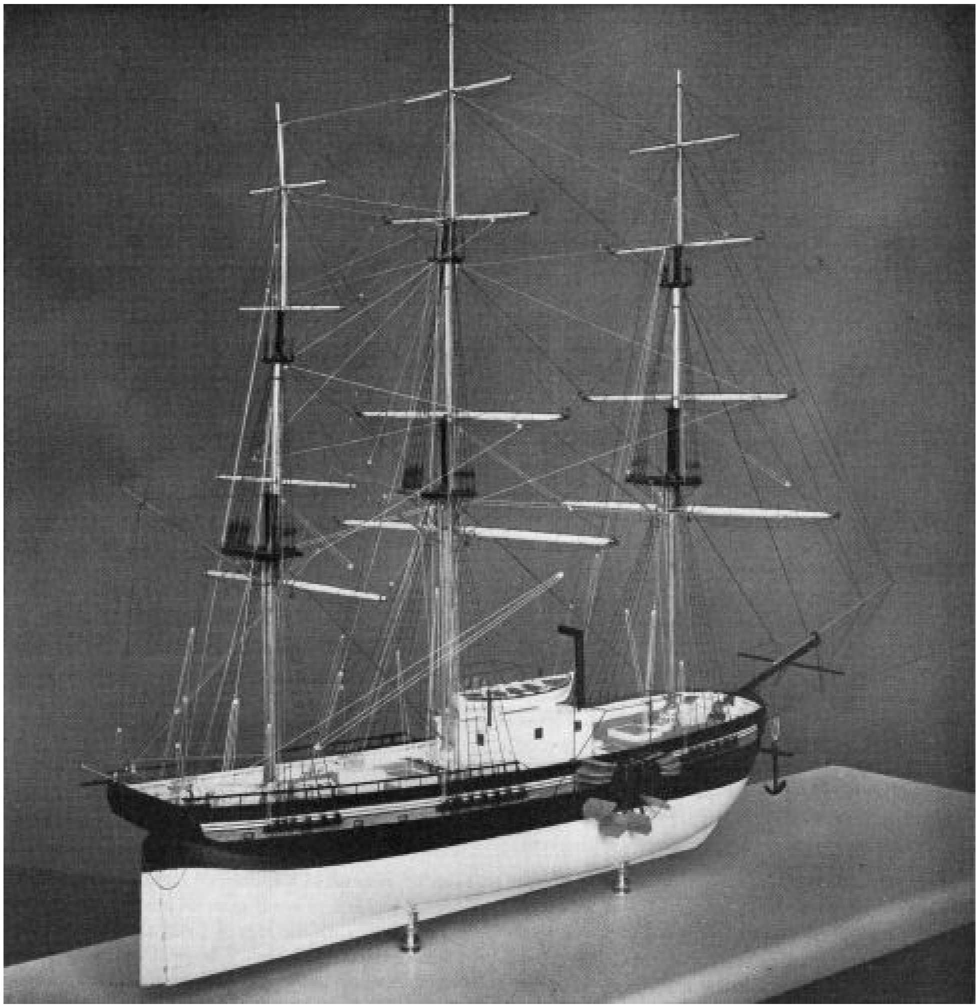

Figure 8.—Stern-quarter view of the new model of the *Savannah,* showing one wheel partially folded and the iron frames for canvas wheel-boxes in place.

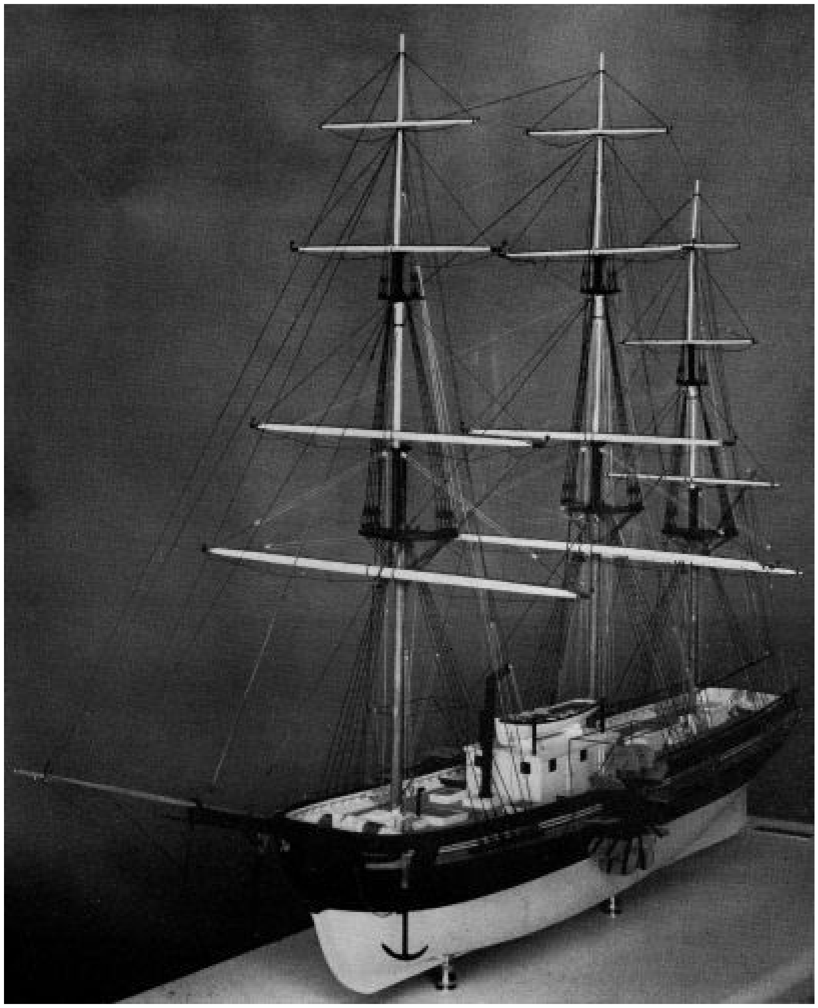

Figure 9.—Bow-quarter view of the new model of the *Savannah*, showing deck arrangement details.

FOOTNOTES:

[1] Carl W. Mitman, *Catalogue of the Watercraft Collection in the United States National Museum*, U.S. National Museum Bulletin 127, 1923.

[2] Jean Baptiste Marestier, *Mémoire sur les Bateaux à Vapeur de Etats-Unis d'Amérique*, Paris, 1823.

[3] A memorandum dated April 20, 1899, in the manuscript file on the watercraft collection shows that the Museum had both the rigged model and the original logbook at that time. Also in the collection were a coffee urn and miniature portrait of the *Savannah's* captain, Moses Rogers, that had been presented to him abroad; later, these items were returned to the donor. A cup and saucer belonging to Captain Rogers also had been given to the Museum, and they are now in its historical collection.

[4] Robert Greenhalgh Albion, *Square Riggers on Schedule*, Princeton, New Jersey, 1938. Between the years 1817 and 1837 the yard of Fickett and Crockett also operated at various times under the name of S. & F. Fickett and the name of Fickett and Thomas. The yard appears to have specialized in the construction of coastal packet ships, because only 4 ocean packets, against 24 coastal packets, were built by the various partnerships in which Fickett was a member.

[5] L. M'Kay, *The Practical Shipbuilder*, New York, 1839.

[6] Howard I. Chapelle, *The Baltimore Clipper*, Salem, Massachusetts, 1930, pp. 112–134.

[7] Sidney Withington, translator, *Memoir on Steamboats of the United States of America by Jean Baptiste Marestier*, Mystic, Connecticut, 1957.

[8] *Ibid.*, pl. 7, figs. 32, 33, 35.

[9] *Ibid.*, pl. 3, fig. 10.

[10] Geo. Henry Preble, *A Chronological History of the Origin and Development of Steam Navigation, 1543–1882*, Philadelphia, 1883.

[11] John H. Morrison, *A History of American Steam Navigation*, New York, 1930.

[12] David Budlong Tyler, *Steam Conquers the Atlantic*, New York and London, 1939.

[13] S. C. Gilfillan, *Inventing the Ship*, New York, 1935.

[14] Charles Frederich Partington, *An Historical and Descriptive Account of the Steam Engine*, London, 1822.

[15] J. Elfreth Watkins, "The Log of the *Savannah*," in *Report of the U.S. National Museum for the Year Ending June 30, 1890*, 1891, pp. 611–639.

[16] Previously, the author had assumed there was one boiler with two flues.

[17] *Op. cit.* (footnote 4).

[18] William Morgan and Augustin Creuze, eds., *Papers on Naval Architecture*, London, n. d., no. 12, p. 387.

[19] Letter from Carl C. Cutler to the author, November 24, 1958.

[20] *Op. cit.* (footnote 5).

[21] Withington, *op. cit.* (footnote 7), pl. 9, figs. 55, 56.

[22] Report of Malcolm Bell, Jr., and Frank Braynard.

[23] M'Kay, *op. cit.* (footnote 5).

[24] Darcy Lever, *Sheet Anchor*, London, 1808–1811. (Reprint, Providence, Rhode Island, 1930.)

[25] David Budlong Tyler, *Steam Conquers the Atlantic*, New York and London, 1939.

[26] S. C. Gilfillan, *Inventing the Ship*, New York, 1935.

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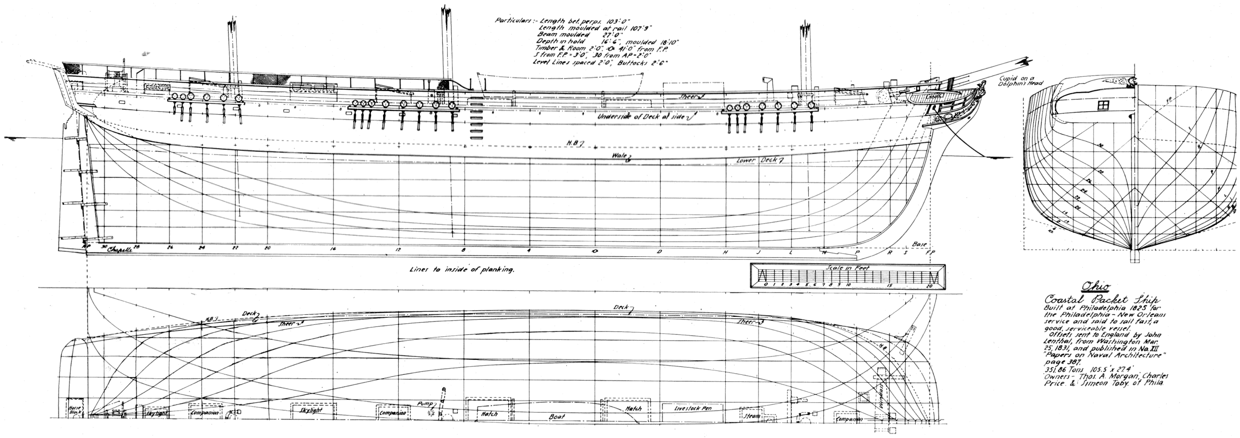
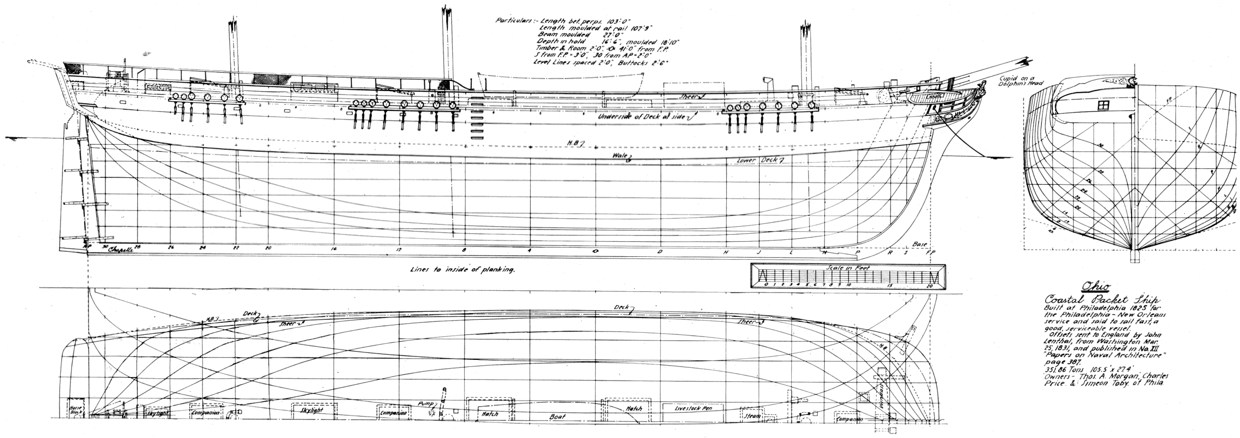

Particulars :- Length bet. perps. 103'0"
Length moulded at rail 107'9"
Beam moulded 27'0"
Depth in hold 16'6", moulded 18'10"
Timber & Room 2'0" to 41'0" from F.P.
5 from F.P. = 3'0", 30 from AP = 2'0"
Level Lines spaced 2'0", Buttocks 2'6"
Underside of Deck at side
Sheer
Wale
Lower Deck
Cupid on a Dolphin's Head
Lines to inside of planking.
Scale in Feet
Ohio
Coastal Packet Ship
Built at Philadelphia 1825 for the Philadelphia - New Orleans service and said to sail fast, a good, serviceable vessel.
Offsets sent to England by John Lenthal, from Washington Mar. 25, 1831, and published in No. XII "Papers on Naval Architecture" page 387.
351.86 Tons 105.5' x 27.4'
Owners - Thos. A. Morgan, Charles Price. & Simeon Toby. of Phila.
Skylight
Companion
Pump
Hatch
Boat
Livestock Pen

back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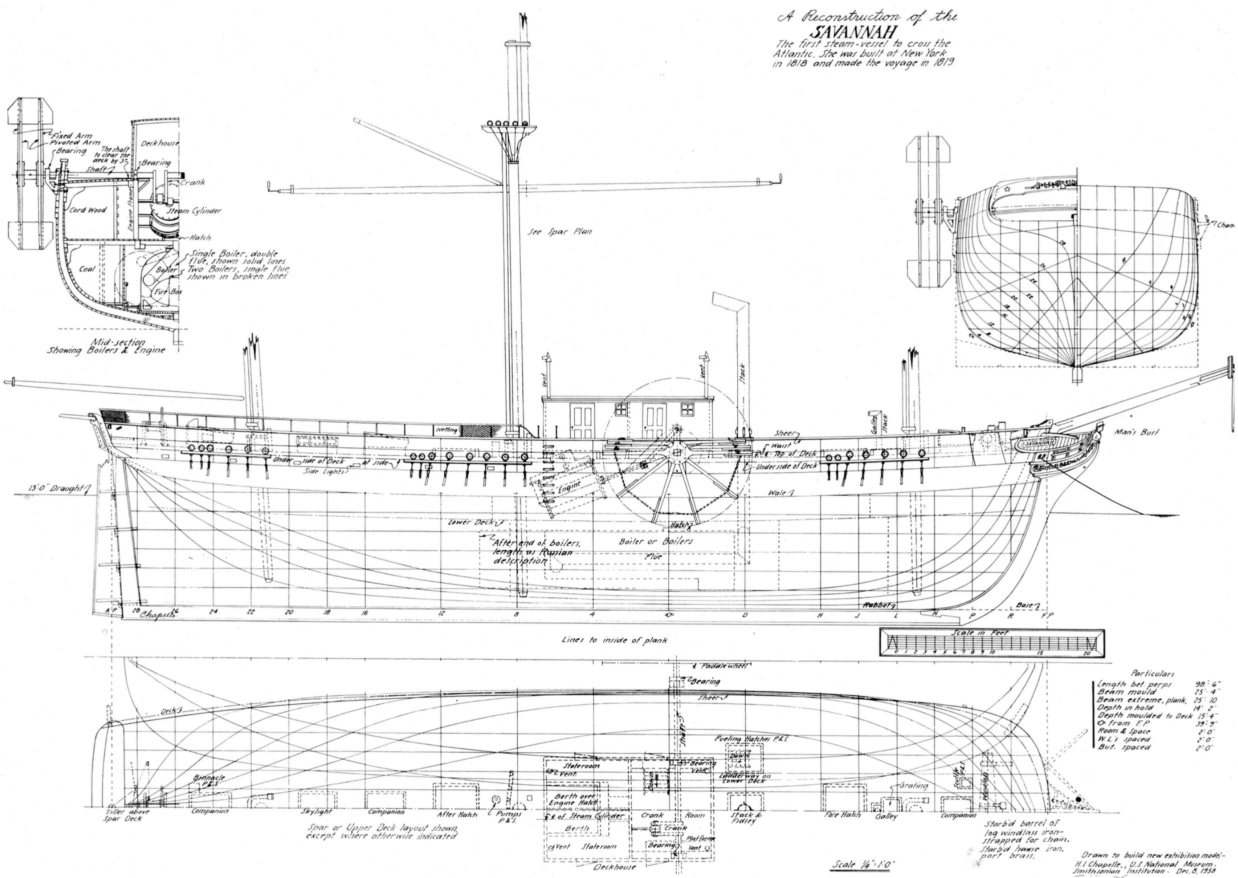
A Reconstruction of the
SAVANNAH
The first steam-vessel to cross the Atlantic. She was built at New York in 1818 and made the voyage in 1819
Mid-section
Showing Boilers & Engine
See Spar Plan
Lines to inside of plank
Scale in Feet
Particulars
Length bet. perps 98' 6"
Beam mould 25' 4"
Beam extreme, plank, 25' 10
Depth in hold 14' 2"
Depth moulded to Deck 15' 4"
from F.P. 39' 9"
Room & Space 2' 0"
W.L.'s spaced 2' 0"
But. spaced 2' 0"
Spar or Upper Deck layout shown, except where otherwise indicated
Scale 1/4" = 1'0"
Drawn to build new exhibition model - H.I. Chapelle., U.S. National Museum. Smithsonian Institution - Dec. 8. 1958

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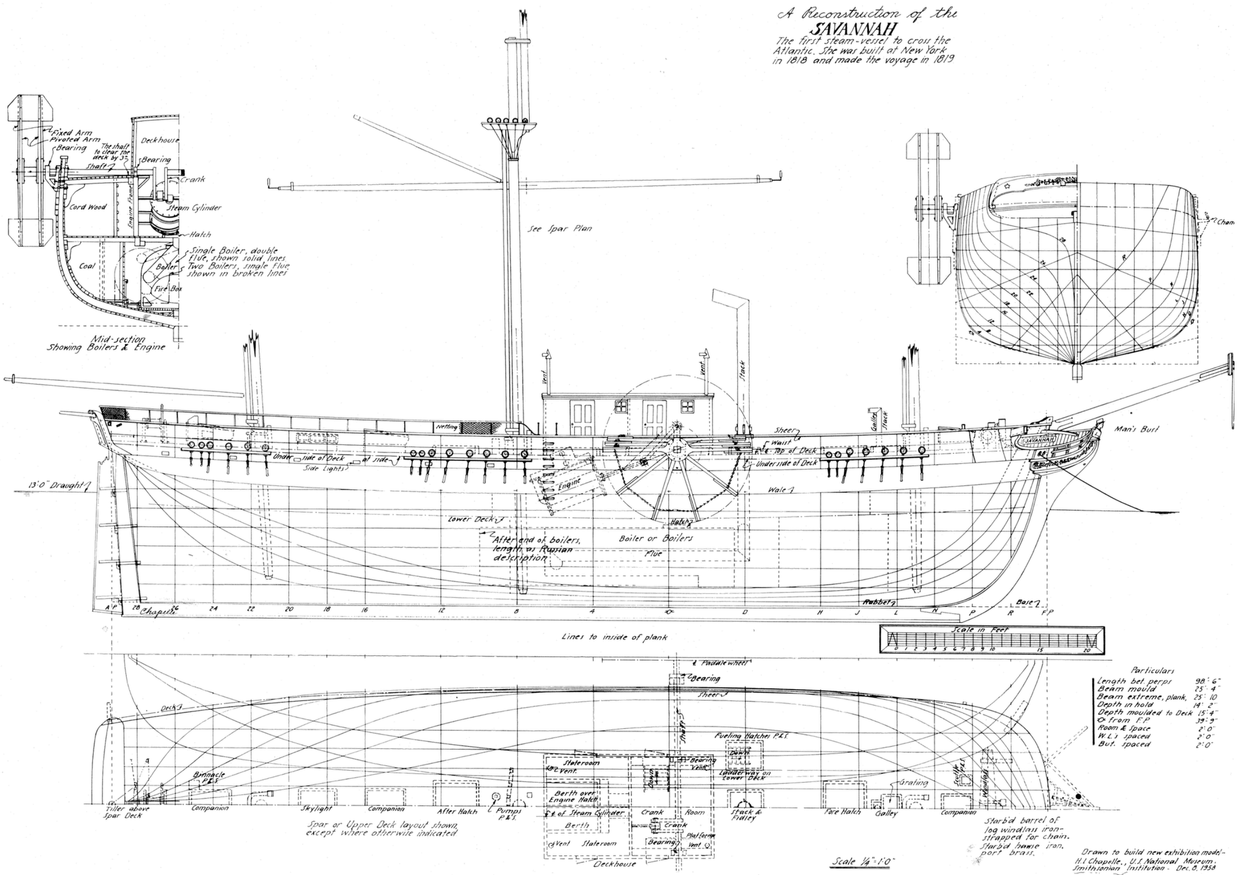
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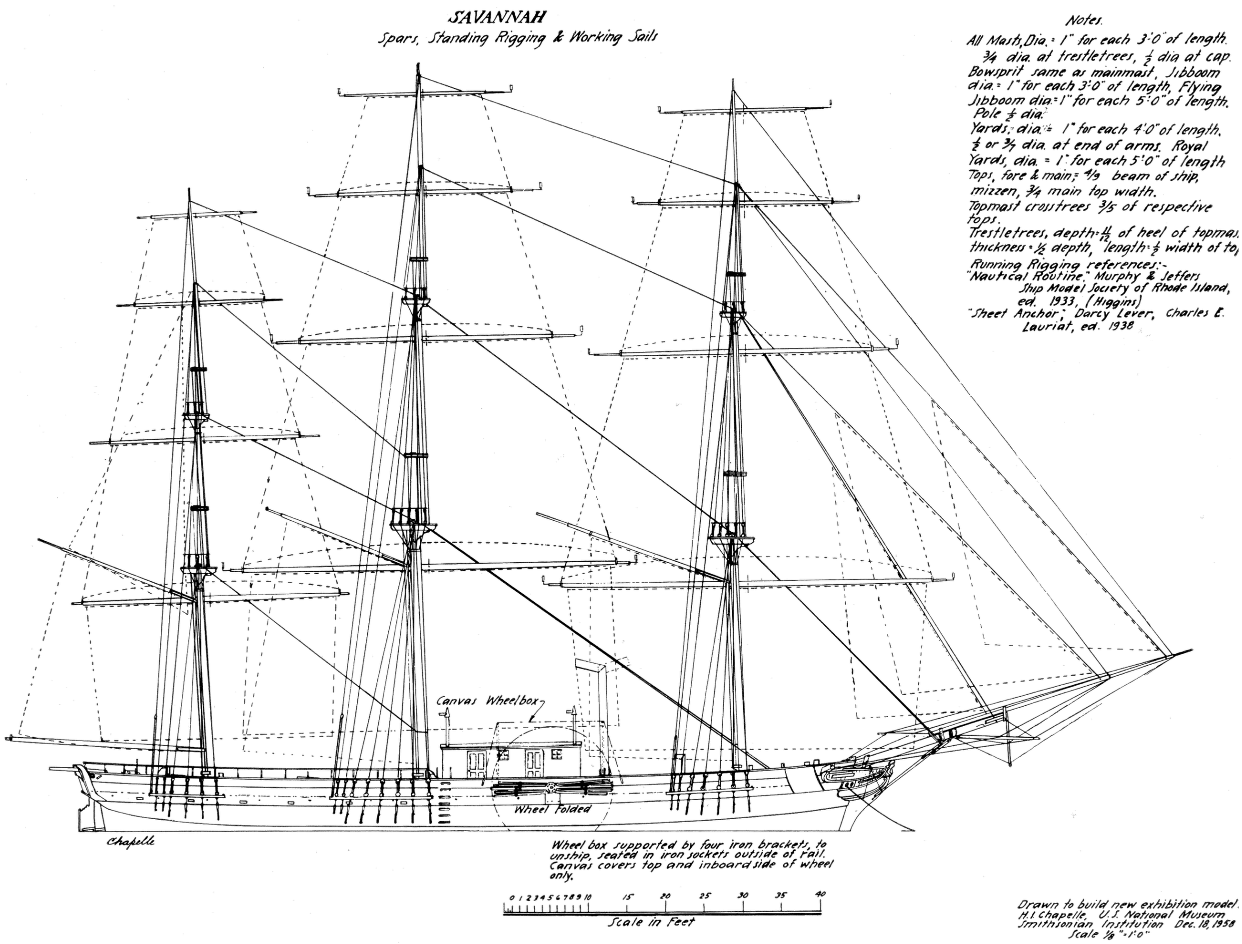
SAVANNAH
Spars, Standing Rigging & Working Sails
Notes.
All Masts, Dia. = 1" for each 3'-0" of length.
3/4 dia. at trestletrees, 1/2 dia at cap.
Bowsprit same as mainmast, Jibboom
dia. = 1" for each 3'-0" of length, Flying
Jibboom dia. = 1" for each 5'-0" of length.
Pole 1/2 dia.
Yards, dia. = 1" for each 4'-0" of length.
1/2 or 3/7 dia. at end of arms. Royal
Yards, dia. = 1" for each 5'-0" of length
Tops, fore & main = 4/9 beam of ship,
mizzen, 3/4 main top width.
Topmast crosstrees 3/5 of respective
tops.
Trestletrees, depth = 11/12 of heel of topmast
thickness = 1/2 depth, length = 1/2 width of top
Running Rigging references:-
"Nautical Routine," Murphy & Jeffers
Ship Model Society of Rhode Island,
ed. 1933, (Higgins)
"Sheet Anchor," Darcy Lever, Charles E.
Lauriat, ed. 1938
Canvas Wheelbox
Wheel Folded
Chapelle
Wheel box supported by four iron brackets, to
unship, seated in iron sockets outside of rail.
Canvas covers top and inboard side of wheel
only.
0 1 2 3 4 5 6 7 8 9 10 15 20 25 30 35 40
Scale in Feet
Drawn to build new exhibition model.
H.I. Chapelle, U.S. National Museum
Smithsonian Institution Dec. 18, 1958
Scale 1/8" = 1'-0"

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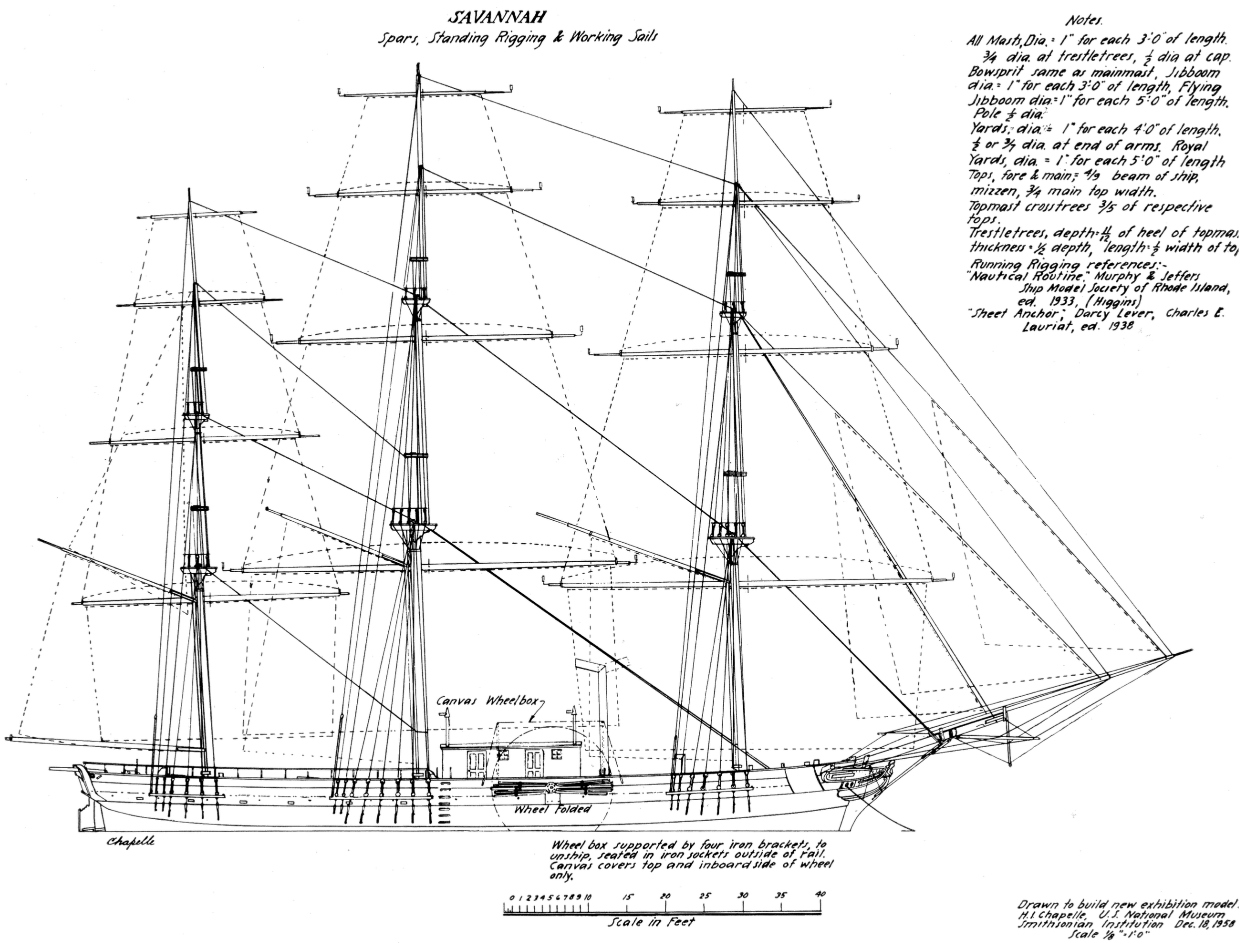
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